

Cost-Sensitive Modeling and Business Impact Analysis

Introduction

Predictive maintenance is an important application of machine learning in industrial systems. It helps detect potential machine failures before they occur, reducing downtime and maintenance costs. In this project, a Random Forest classification model was used to predict machine failures using operational parameters such as temperature, rotational speed, torque, and tool wear.

However, the dataset contains a **class imbalance problem**, where machine failure cases are much fewer compared to normal machine conditions. Therefore, evaluating the model based only on accuracy may not be sufficient. This report focuses on analyzing the model performance, understanding the cost of prediction errors, and estimating the potential business loss.

Model Evaluation

Accuracy

The model achieved an overall **accuracy of 98.15%**, indicating that most machine conditions were correctly predicted.

Confusion Matrix

Actual / Predicted	Normal	Failure
Normal	1935	4
Failure	33	28

Interpretation:

- **True Negatives (1935)** : Normal machines correctly predicted as normal.
- **True Positives (28)** : Failed machines correctly detected.
- **False Positives (4)** : Normal machines incorrectly predicted as failures.
- **False Negatives (33)** : Failed machines incorrectly predicted as normal.

False negatives are critical in predictive maintenance because missing a failure can lead to unexpected machine breakdown and production loss.

Recall for Minority Class

The recall for the machine failure class is **0.46**. This means the model correctly detects **46% of the actual machine failures** while some failures are still missed.

Improving recall is important in predictive maintenance because early detection of failures can prevent costly downtime and damage.

Precision Trade-off

The precision for the failure class is **0.88**, which means that when the model predicts a machine failure, it is correct **88% of the time**.

However, increasing recall may sometimes reduce precision, resulting in more false alarms. Therefore, there is a trade-off between detecting more failures and avoiding unnecessary maintenance operations.

Cost Matrix Definition

Actual / Predicted	Normal	Failure
Normal	0	₹5000
Failure	₹80000	0

Cost of Missed Failure (False Negative): ₹80,000

If the model fails to detect a machine failure, the consequences may include:

- Unexpected machine breakdown
- Production downtime
- Repair costs
- Labor and operational losses

Therefore, the cost of missing a failure is considered very high.

Cost of False Alarm (False Positive): ₹5,000

If the model predicts a failure when the machine is actually normal, it may result in:

- Unnecessary inspection
- Preventive maintenance
- Temporary operational interruption

Although this causes some cost, it is significantly lower than the cost of missing a failure.

- Missing a machine failure (False Negative) is very expensive because it can lead to machine breakdown and production loss.
- A false alarm (False Positive) only causes unnecessary inspection or maintenance.

Expected Business Loss Calculation

Using the confusion matrix results:

- False Negatives (FN) = 33
- False Positives (FP) = 4

Cost assumptions:

- Cost of missed failure = ₹80,000
- Cost of false alarm = ₹5,000

Business Loss Formula

Expected Loss = (FN × Cost of Missed Failure) + (FP × Cost of False Alarm)

Calculation:

$$33 \times 80,000 = ₹26,40,000$$

$$4 \times 5,000 = ₹20,000$$

Total Expected Business Loss : **₹26,60,000**

Based on the confusion matrix, the model produced **33 false negatives and 4 false positives**. Considering the operational cost of ₹80,000 for each missed failure and ₹5,000 for each false alarm, the expected business loss is approximately **₹26,60,000**. This shows that missed failures contribute the most to business loss.

Conclusion

The machine learning model achieved high accuracy, but the class imbalance problem caused several machine failures to remain undetected. Although the number of false alarms was low, the missed failures contributed significantly to the overall business loss.

This analysis highlights the importance of using **cost-sensitive modeling and imbalance handling techniques such as SMOTE or class weighting** to improve the detection of machine failures.

Improving recall for the minority class can significantly reduce operational risk and financial losses in predictive maintenance systems.

Business Cost Comparison Table

Model	False Negatives (FN)	False Positives (FP)	FN Cost (₹80,000)	FP Cost (₹5,000)	Total Business Loss
Random Forest (Baseline Model)	33	4	₹26,40,000	₹20,000	₹26,60,000
Random Forest (Class Weight Model)	33	4	₹26,40,000	₹20,000	₹26,60,000
Random Forest (SMOTE Model)	27	58	₹21,60,000	₹2,90,000	₹24,50,000

Conclusion

The SMOTE-based Random Forest model reduced the number of false negatives from 33 to 27. Since missed failures have a higher operational cost, the overall business loss decreased from ₹26,60,000 to ₹24,50,000. Therefore, the SMOTE model provides a better cost-sensitive solution compared to the baseline model.